XGBoost, and Comparing Trees, Bagging and Boosting

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Sources: `Lectures/L12_student_trees_bagging_boosting_worked_deck.pdf` (Part 5, slides 82–96; Part 6, slides 97–102) · Previous: AdaBoost · Next: Honest evaluation

Lecture 12 files: L12a Trees → L12b Bagging & random forests → L12c Gradient boosting → L12d AdaBoost → **L12e XGBoost & comparison**

> **How to use this file.** XGBoost is the gradient boosting of L12c with three upgrades: (1) it writes one **objective** with slope g and curvature h and solves it exactly per leaf; (2) it adds a penalty **λ** on large leaf values; (3) it adds a penalty **γ** per leaf, so a split must *earn* its place. The trees can now be deeper than stumps, growing while the gain beats γ. Part A builds this step by step on the twelve runs; A8–A10 compare every method of Lecture 12. Part B walks slides 82–102. Part C is practice.

Setting used throughout: λ = 1, γ = 0.25, step ν = 0.5, trees at most two questions deep. Twelve runs from L12a ($y = 1$ for D, 0 for C); new runs **A** (0.05, 32, no) and **B** (0.2, 64, yes).

> **Careful. Symbol warning.** L12c called the leaf value γ. XGBoost calls the leaf value **w** and uses **γ for the penalty per leaf**.

What this file is really about:

$$g_i = p_i - y_i,\ h_i = p_i(1-p_i) \ \rightarrow\ w_j = -\frac{G_j}{H_j + \lambda} \ \rightarrow\ \text{gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}\right] - \gamma \ \rightarrow\ F \leftarrow F$$

Step-up path

You need. Gradient boosting and a second-order Taylor expansion.

1. Approximate the loss with slopes and curvatures.
2. Add penalties, and solve for the best leaf weight.
3. Score a split by what it saves, minus a price.
4. Compare all the ensembles on what they reduce and what they risk.

You can stop when you can compute a gain from G, H, λ and γ, and say what each penalty does.

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Part A Maths From the Ground Up

A1 What XGBoost changes

Same loop as L12c: current probabilities → fit a tree to what is left → add a shrunken correction. With $g_i = p_i - y_i = -r_i$ (slope), $h_i = p_i(1-p_i)$ (curvature), and sums $G = \sum g_i$, $H = \sum h_i$ over a node:

	gradient boosting (L12c)	XGBoost
question chosen by	largest $n_L \bar{r}_L^2 + n_R \bar{r}_R^2$	largest gain built from $G^2/(H+\lambda)$ of yes, no and parent, minus γ
leaf value	$\sum r / \sum p(1-p) = -G/H$	$w = -G/(H+\lambda)$
penalty on leaf values	none	λ: larger λ ⇒ smaller $\lvert w \rvert$
penalty on tree size	none	γ: a split must gain more than γ
tree	a stump	deeper, while a split gains more than γ
update	$F \leftarrow F + \nu\gamma_{\text{leaf}}$	$F \leftarrow F + \nu w_{\text{leaf}}$

A2 The objective and the leaf weight

Step 1 — Second-order loss of one run when its score moves by w (L12c A8): $L(F_i + w) \approx L(F_i) + g_i w + \frac{1}{2} h_i w^2$.

Step 2 — Add the penalties. A new tree with T leaves and leaf weights $w_1, \ldots, w_T$ (run i gets $w_{j(i)}$):

$$\text{Obj} = \sum_i \left[g_i w_{j(i)} + \tfrac{1}{2} h_i w_{j(i)}^2 \right] + \gamma T + \tfrac{1}{2}\lambda \sum_j w_j^2 .$$

Step 3 — Group by leaf. With G_j, H_j the sums in leaf j:

$$\text{Obj} = \sum_{j=1}^{T} \left[G_j w_j + \tfrac{1}{2}(H_j + \lambda) w_j^2 \right] + \gamma T .$$

Each leaf is a **separate parabola** in w_j.

Step 4 — Minimise each: $G_j + (H_j + \lambda) w_j = 0 \Rightarrow$

$$\boxed{w_j = -\frac{G_j}{H_j + \lambda}}, \qquad \text{leaf contributes } -\tfrac{1}{2}\frac{G_j^2}{H_j + \lambda} \text{ to the objective.}$$

(Plug in: $G \cdot (-\frac{G}{H+\lambda}) + \frac{1}{2}(H+\lambda)\frac{G^2}{(H+\lambda)^2} = -\frac{1}{2}\frac{G^2}{H+\lambda}$.)

Step 5 — Check against L12c. λ = 0 ⇒ $w = -G/H = \sum r / \sum p(1-p)$, the Newton leaf value. λ in the denominator pulls every leaf **towards 0**, by the factor $H/(H+\lambda)$ — most strongly when the leaf is **small** (H small). That is ridge regularisation (L07) applied to leaf values.

A3 The gain of a split

Step 1 — Before: a node with (G, H) as one leaf: $-\frac{1}{2}\frac{G^2}{H+\lambda} + \gamma$.

Step 2 — After: split into (G_L, H_L) and (G_R, H_R): $-\frac{1}{2}\frac{G_L^2}{H_L+\lambda} - \frac{1}{2}\frac{G_R^2}{H_R+\lambda} + 2\gamma$.

Step 3 — Fall in the objective:

$$\boxed{\text{gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda}\right] - \gamma}$$

> **Rule:** take the question with the largest gain; if it is ≤ 0, don't split (make a leaf).

Step 4 — Compare with L12a's entropy gain. Same shape: parent vs its two groups. Here the "purity" of a group is $G^2/(H+\lambda)$: large when the runs all push the **same way** (their g_i share a sign), zero when they cancel. And with λ = γ = 0 and all h equal, $G^2/H \propto (\sum r)^2/n$ — L12c's score.

A4 Round 1: slopes, curvatures, the root question

Step 1 — Start: all $F = 0 \Rightarrow p = 0.5$.

runs	y	$g = p - y$	$h = p(1-p)$	how many
1–6 (D)	1	−0.5	0.25	6
7–12 (C)	0	+0.5	0.25	6

Step 2 — Parent: $G = 6(-0.5) + 6(0.5) = 0$, $H = 12 \times 0.25 = 3 \Rightarrow G^2/(H+\lambda) = 0/4 = 0$.

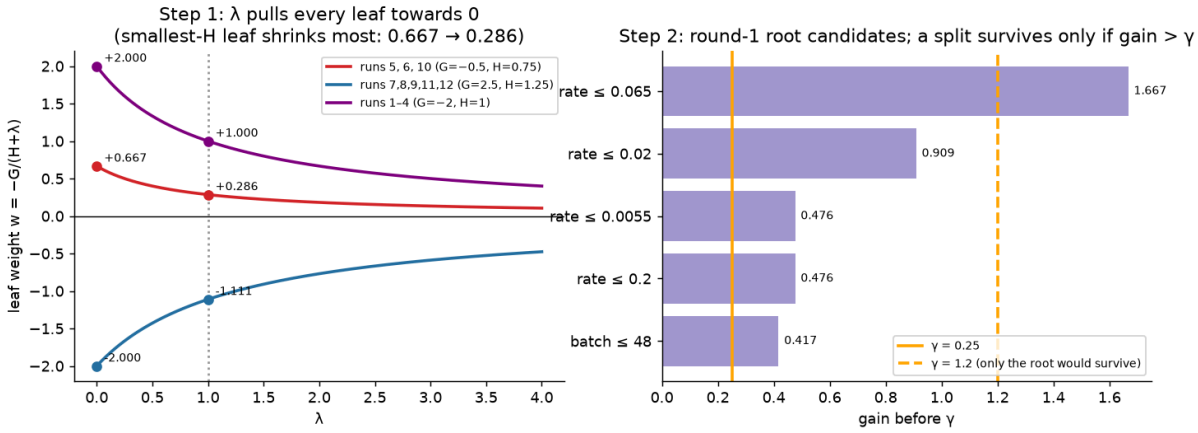
Step 3 — Every candidate:

question	yes-group (G, H)	no-group (G, H)	gain (before γ)
rate ≤ 0.065	+2.00, 2.00	−2.00, 1.00	**1.667**
rate ≤ 0.02	+1.50, 1.25	−1.50, 1.75	0.909
rate ≤ 0.0055	+1.00, 0.50	−1.00, 2.50	0.476
rate ≤ 0.2	+1.00, 2.50	−1.00, 0.50	0.476
batch ≤ 48	−1.00, 1.00	+1.00, 2.00	0.417

Step 4 — Winner in full:

$$\text{gain} = \tfrac{1}{2}\left[\frac{2^2}{2+1} + \frac{(-2)^2}{1+1} - \frac{0^2}{3+1}\right] = \tfrac{1}{2}[1.333 + 2.000 - 0] = 1.667 > \gamma = 0.25 \;\Rightarrow\; \text{split}.$$

(Yes-group $G = +2$: six C runs push up the slope, two D runs down — net: "lower these scores".)



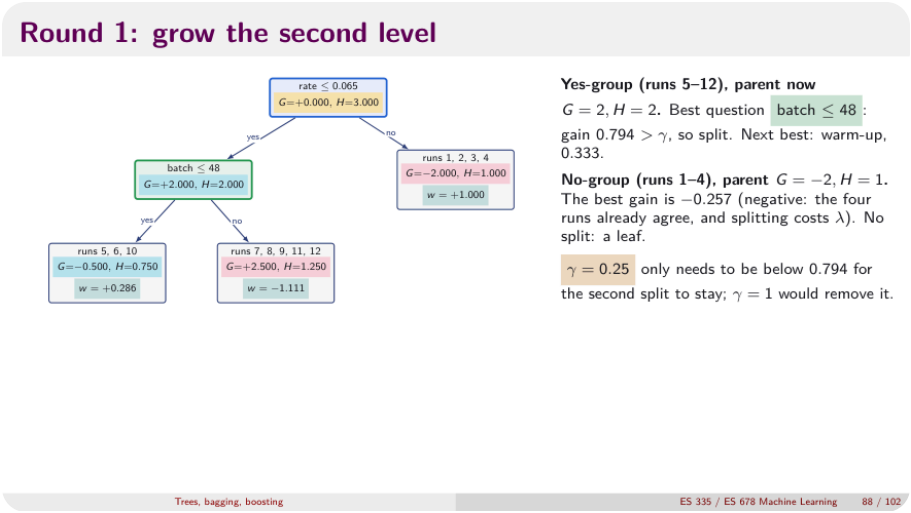
Step 1: λ shrinks every leaf; the smallest-H leaf shrinks most. Step 2: a split survives only if its gain clears γ.

A5 Round 1: second level, leaf weights, update

Step 1 — Yes-group (runs 5–12) is now a parent with $G = 2$, $H = 2$. Best question **batch ≤ 48**: gain **0.794 > γ** ⇒ split (next best warm-up, 0.333).

Step 2 — No-group (runs 1–4), $G = -2$, $H = 1$. Best gain **−0.257**: negative! The four runs already agree, and splitting costs λ (each smaller leaf is shrunk more). ⇒ leaf.

Step 3 — γ's role here. γ = 0.25 only needs to be below 0.794 for the second split to stay; γ = 1 would remove it.



Slide 88 — A depth-2 tree with three leaves.

Step 4 — Leaf weights $w = -G/(H+1)$:

leaf	G	H	w (λ = 1)	w (λ = 0)
runs 5, 6, 10	−0.500	0.750	**+0.286**	+0.667
runs 7, 8, 9, 11, 12	+2.500	1.250	**−1.111**	−2.000
runs 1, 2, 3, 4	−2.000	1.000	**+1.000**	+2.000

Every value is smaller than the Newton value by $H/(H+\lambda)$; the leaf with the fewest runs and least curvature (5, 6, 10; $H = 0.75$) is shrunk most: 0.286 vs 0.667.

Step 5 — Update $F \leftarrow F + 0.5w$:

runs	leaf	vw	F after	p after
1–4 (D)	(1, 2, 3, 4)	+0.500	+0.500	0.622
5, 6 (D)	(5, 6, 10)	+0.143	+0.143	0.536
10 (C)	(5, 6, 10)	+0.143	+0.143	0.536 ×
7, 8, 9, 11, 12 (C)	(7, 8, 9, 11, 12)	−0.556	−0.556	0.365

Log loss 0.693 → **0.515** (L12c's stump: 0.504). Only run 10 is wrong — it shares a leaf with 5 and 6.

"Your turn" (slides 90–91). Node $G = +1$, $H = 2$ splits into $(G_L, H_L) = (-1, 1)$, $(G_R, H_R) = (2, 1)$, λ = 1.

1. $\text{gain} = \frac{1}{2}\left[\frac{1}{2} + \frac{4}{2} - \frac{1}{3}\right] = \frac{1}{2}[0.5 + 2 - 0.333] = 1.083.$
2. $w_L = -(-1)/2 = +0.5$, $w_R = -2/2 = -1.0$ (unsplit: $-1/3$).
3. γ = 0.25: $1.083 - 0.25 = 0.833 > 0$ ⇒ keep. γ = 1.2: < 0 ⇒ no split. **Larger γ ⇒ smaller trees.**

A6 Rounds 1–8, and what λ and γ do

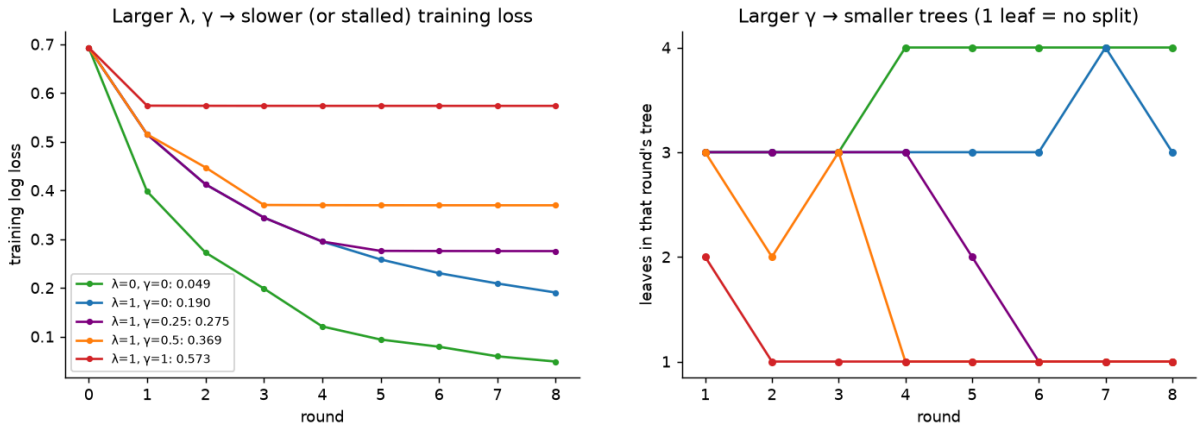
Step 1 — Rounds 1–6 (λ = 1, γ = 0.25):

round	questions (root, then yes-side)	gains before γ	leaf weights w	loss	wrong
1	rate ≤ 0.065, then batch ≤ 48	1.667, 0.794	+0.29, −1.11, +1.00	0.515	10
2	rate ≤ 0.065, then batch ≤ 48	0.939, 0.462	+0.23, −0.84, +0.78	0.412	10
3	rate ≤ 0.065, then warm-up = no	0.575, 0.343	+0.17, −0.75, +0.64	0.344	none
4	rate ≤ 0.065, then batch ≤ 48	0.379, 0.282	+0.21, −0.63, +0.54	0.295	none
5	rate ≤ 0.065	0.256	−0.26, +0.46	0.276	none
6	no split	all below γ	+0.052	0.276	none

Training error 0 from round 3. In round 5 only the root clears γ; from round 6 nothing gains more than 0.25, so the "tree" is a single leaf that shifts every score by 0.03, 0.02, 0.01 and the loss stalls near 0.275. **γ stopped the learning by itself.**

Step 2 — Five settings, eight rounds:

settings	first round with no wrong run	wrong after 8	log loss after 8	leaves in trees 1–8
λ = 0, γ = 0	2	none	0.049	3, 4, 4, 4, 4, 4, 4, 4
λ = 1, γ = 0	3	none	0.190	3, 3, 3, 3, 3, 3, 4, 3
λ = 1, γ = 0.25	3	none	0.275	3, 3, 3, 3, 2, 1, 1, 1
λ = 1, γ = 0.5	never	10	0.369	3, 2, 3, 1, 1, 1, 1, 1
λ = 1, γ = 1	never	5, 6	0.573	2, 1, 1, 1, 1, 1, 1, 1



Larger λ and γ ⇒ smaller leaf values and smaller trees ⇒ training loss falls more slowly or stops. That is the intended price of a simpler model; too large γ (≥ 0.5 here) stops before the training data are even fitted. **Choose both on held-out data.**

A7 Runs A and B; the XGBoost recipe

Step 1 — Paths (entry = vw of the leaf reached):

round	run A (0.05, 32, no): path	vw	run B (0.2, 64, yes): path	vw
1	rate yes; batch ≤ 48 yes	+0.143	rate ≤ 0.065 no	+0.500
2	rate yes; batch ≤ 48 yes	+0.113	rate no	+0.389
3	rate yes; warm-up = no yes	+0.086	rate no	+0.319
4	rate yes; batch ≤ 48 yes	+0.104	rate no	+0.269
5	rate yes	−0.131	rate no	+0.232

Rounds 6–8 add +0.026, +0.017, +0.012 to every run. Totals: **A: F = 0.369, p = 0.591**; **B: F = 1.764, p = 0.854**. Both D; B with more confidence. Unlike AdaBoost, XGBoost gives a **probability**.

Step 2 — The recipe:

step	what you do	which number is used
start	$F_i = 0$, $p_i = 0.5$; choose λ, γ, ν, depth	the settings
(a) slopes	$g_i = p_i - y_i$, $h_i = p_i(1 - p_i)$	each run's current p
(b) question	largest gain $\frac{1}{2}[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}] - \gamma$	G, H of node, yes, no
(c) grow	split again while gain > 0; stop at the depth limit	gain after γ
(d) leaf weights	$w = -G/(H + \lambda)$	G, H of the leaf, λ
(e) update	$F_i \leftarrow F_i + \nu w$	w of the leaf reached
predict	$F = \nu \sum_t w_t(\text{leaf})$, $p = S(F)$	one weight per tree

A8 Comparison: bagging vs boosting

	bagging (random forest)	boosting
trees grown	side by side, each on its own bootstrap sample	one after another, each on the residuals so far
each tree	deep: low bias, high variance	small (a stump): high bias, low variance
combined by	average $\frac{1}{B}\sum_b T_b$	sum $F_0 + \nu \sum_t \gamma_t(\cdot)$
mainly reduces	variance, down to the floor ρσ²	bias, step by step
more trees	error settles; harmless	keeps fitting; stop using held-out data
main settings	number of trees; features per split	step size; rounds; depth

Numbers behind this: L12b A12 (bias² 5.25 → 0.011 for boosting; variance 0.252 → 0.146 for bagging).

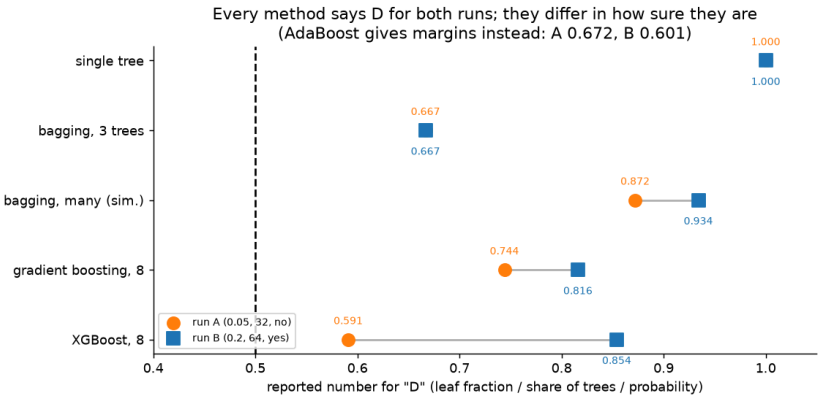
A9 Comparison: gradient boosting, AdaBoost, XGBoost

	gradient boosting (L12c)	AdaBoost (L12d)	XGBoost (this file)
each stump/tree is fitted to	residuals $y - p$	the data with run weights	slopes g and curvatures h
loss reduced	log loss	exponential loss e^{-yF}	log loss, plus penalties
question chosen by	$n_L \bar{r}_L^2 + n_R \bar{r}_R^2$	smallest weighted error ε	largest gain with $G^2/(H+\lambda)$
number each tree contributes	$\nu\gamma_{\text{leaf}}$	$\alpha h(x) = \pm\alpha$	νw_{leaf}
output	probability $S(F)$	class and margin F	probability $S(F)$
controls on size	step size, rounds	rounds	λ, γ, depth, step size

One family. All three are *stagewise additive models*: $F \leftarrow F + (\text{small tree})$, the tree chosen to reduce a loss using its gradient. They differ only in the loss (log vs exponential), how the mistakes are passed on (residuals, weights, slope + curvature) and the regularisation.

A10 All methods on runs A and B

method	run A (0.05, 32, no)	run B (0.2, 64, yes)
single tree	D (2/2 in its leaf)	D (4/4 in its leaf)
bagging, 3 trees	0, 1, 1 → 2/3, D	1, 0, 1 → 2/3, D
bagging, many trees (simulated)	≈ 0.872, D	≈ 0.934, D
gradient boosting, 8 stumps	F = 1.069, p = 0.744, D	F = 1.490, p = 0.816, D
AdaBoost, 5 stumps	F = 0.672 (margin), D	F = 0.601 (margin), D
XGBoost, 8 trees	F = 0.369, p = 0.591, D	F = 1.764, p = 0.854, D



Every method says D for both; they differ in how sure. The single tree gives only a class (with false certainty); bagging shows how much the trees disagree (about 0.9 say D); boosting builds a probability from summed corrections. Run A — the "runs 5 and 6 corner" — is consistently the less certain one (except for the single tree, which cannot express doubt).

Part B Slide by Slide (slides 82–102)

Slide 82 Part 5: XGBoost (divider)

Colours: parent node, yes-group, no-group, setting λ, γ (orange), leaf weight w (teal).

Slide 83 What XGBoost changes

What XGBoost changes

Same loop as Part 3: compute each run's current probability, fit a tree to what is left, add a shrunken correction. With $g_i = p_i - y_i = -r_i$ (slope), $h_i = p_i(1-p_i)$ (curvature), and sums $G = \sum g_i$, $H = \sum h_i$ over a node:

	gradient boosting (Part 3)	XGBoost
question chosen by	largest $n_L \bar{r}_L^2 + n_R \bar{r}_R^2$	largest gain, built from $G^2/(H+\lambda)$ of the yes, no and parent nodes, minus γ (formula on a later slide)
leaf value	$\sum r / \sum p(1-p) = -G/H$	$w = -G/(H+\lambda)$
penalty on leaf values	none	λ: larger λ gives smaller $\lvert w \rvert$
penalty on tree size	none	γ: a split must gain more than γ
tree	a stump	deeper, while a split gains more than γ
update	$F \leftarrow F + \nu\gamma_{\text{leaf}}$	$F \leftarrow F + \nu w_{\text{leaf}}$

Symbol warning: Part 3 called the leaf value γ. XGBoost calls it w and uses γ for the penalty per leaf.

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Slide 83: what XGBoost changes from gradient boosting. Slopes and curvatures choose the question, and penalties shrink the leaves.

Slide 84 The objective and the leaf weight

The objective and the leaf weight

For a new tree with T leaves and leaf weights $w_1, \ldots, w_T$ (run i receives $w_{j(i)}$), XGBoost minimises, to second order in the correction,

$$\text{Obj} = \sum_i \left[g_i w_{j(i)} + \tfrac{1}{2} h_i w_{j(i)}^2 \right] + \gamma T + \tfrac{1}{2} \lambda \sum_j w_j^2 = \sum_{j=1}^{T} \left[G_j w_j + \tfrac{1}{2} (H_j + \lambda) w_j^2 \right] + \gamma T,$$

where G_j, H_j add g_i, h_i over the runs in leaf j. Each leaf is a separate quadratic in w_j. Set its derivative to zero:

$$G_j + (H_j + \lambda) w_j = 0 \implies \boxed{w_j = -\frac{G_j}{H_j + \lambda}}, \qquad \text{leaf contributes } -\tfrac{1}{2} \frac{G_j^2}{H_j + \lambda}.$$

Check against Part 3. With $\lambda = 0$: $w = -G/H = \sum r / \sum p(1-p)$, the Newton leaf value already used. The new ingredient is λ in the denominator: it pulls every leaf value towards 0, and more strongly when the leaf is small (H small).

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Slide 84 — λ pulls every leaf towards 0, more so when H is small (A2).

Slide 85 The gain of a split

The gain of a split

A node with sums (G, H) is split into (G_L, H_L) and (G_R, H_R). The objective before is $-\frac{1}{2}\frac{G^2}{H+\lambda}+\gamma$ (one leaf); after, $-\frac{1}{2}\frac{G_L^2}{H_L+\lambda}-\frac{1}{2}\frac{G_R^2}{H_R+\lambda}+2\gamma$ (two leaves). The fall in the objective is

$$\text{gain}=\frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda}+\frac{G_R^2}{H_R+\lambda}-\frac{G^2}{H+\lambda}\right]-\gamma$$

Rule: among all questions take the largest gain; if it is ≤ 0 do not split (make a leaf).

Compare with Part 1. The entropy gain also compared the parent (parent) with its two groups (yes , no). Here the "impurity" of a group is $G^2/(H+\lambda)$: large when the runs in the group all push the same way (their g_i share a sign), zero when they cancel.

Slide 85 — "Impurity" of a group is $G^2/(H+\lambda)$ (A3).

Slide 86 Round 1: gradients and curvatures

Round 1: gradients and curvatures

Settings used from here on: $\lambda=1$, $\gamma=0.25$, step size $\nu=0.5$, trees at most two questions deep. All scores start at 0, so every $p=0.5$.

runs	y	$g=p-y$	$h=p(1-p)$	how many
1–6 (**D**)	1	-0.5	0.25	6
7–12 (**C**)	0	$+0.5$	0.25	6

Whole data set, the parent node: $G=6(-0.5)+6(0.5)=0$, $H=12\times 0.25=3$, so $\frac{G^2}{H+\lambda}=\frac{0^2}{3+1}=0$.

Slide 86: the settings for round one, each run's slope and curvature, and the sums for the parent node.

Slide 87 Round 1: choose the root question

Round 1: choose the root question

Every candidate question, with (G, H) of each group:

question	yes-group (G, H)	no-group (G, H)	gain (before γ)
rate ≤ 0.065	$+2.00,\ 2.00$	$-2.00,\ 1.00$	1.667
rate ≤ 0.02	$+1.50,\ 1.25$	$-1.50,\ 1.75$	0.909
rate ≤ 0.0055	$+1.00,\ 0.50$	$-1.00,\ 2.50$	0.476
rate ≤ 0.2	$+1.00,\ 2.50$	$-1.00,\ 0.50$	0.476
batch ≤ 48	$-1.00,\ 1.00$	$+1.00,\ 2.00$	0.417

Winner: rate ≤ 0.065.

$$\text{gain} = \tfrac{1}{2}\left[\frac{2^2}{2+1} + \frac{(-2)^2}{1+1} - \frac{0^2}{3+1}\right] = \tfrac{1}{2}\left[1.333 + 2.000 - 0\right] = \mathbf{1.667} > \gamma = 0.25$$

Slide 87: every candidate root question with the slopes and curvatures of its two groups, and its gain. The rate ≤ 0.065 question wins.

Slide 88 Round 1: grow the second level

Shown in A5. Yes side: batch ≤ 48, gain 0.794. No side: best gain −0.257 ⇒ leaf.

Slide 89 Round 1: leaf weights

Round 1: leaf weights

$$w_j = -\frac{G_j}{H_j + \lambda}, \qquad \lambda = 1$$

leaf	G	H	w with $\lambda = 1$	w with $\lambda = 0$
runs 5, 6, 10	-0.500	0.750	$+0.286$	$+0.667$
runs 7, 8, 9, 11, 12	$+2.500$	1.250	-1.111	-2.000
runs 1, 2, 3, 4	-2.000	1.000	$+1.000$	$+2.000$

Every leaf value is smaller than the Part 3 (Newton) value, by the factor $H/(H + \lambda)$. The leaf with the fewest runs and the least curvature (runs 5, 6, 10, $H = 0.75$) is shrunk the most: 0.286 against 0.667. Update: $F \leftarrow F + \nu w$ with $\nu = 0.5$.

Slide 89: each leaf's weight with and without the penalty. Every weight is smaller with λ = 1, by the factor H/(H+λ).

Slides 90–91 Your turn: is this split worth it? (and answer)

Gain 1.083; $w_L = +0.5$, $w_R = -1.0$; keep at γ = 0.25, reject at γ = 1.2 (A5).

Slide 92 Round 1: update every run

Round 1: update every run

run	out	p before	g	leaf (runs)	w	νw	F after	p after
1	D	0.500	−0.500	(1, 2, 3, 4)	+1.000	+0.500	+0.500	0.622
2	D	0.500	−0.500	(1, 2, 3, 4)	+1.000	+0.500	+0.500	0.622
3	D	0.500	−0.500	(1, 2, 3, 4)	+1.000	+0.500	+0.500	0.622
4	D	0.500	−0.500	(1, 2, 3, 4)	+1.000	+0.500	+0.500	0.622
5	D	0.500	−0.500	(5, 6, 10)	+0.286	+0.143	+0.143	0.536
6	D	0.500	−0.500	(5, 6, 10)	+0.286	+0.143	+0.143	0.536
7	C	0.500	+0.500	(7, 8, 9, 11, 12)	−1.111	−0.556	−0.556	0.365
8	C	0.500	+0.500	(7, 8, 9, 11, 12)	−1.111	−0.556	−0.556	0.365
9	C	0.500	+0.500	(7, 8, 9, 11, 12)	−1.111	−0.556	−0.556	0.365
10	C	0.500	+0.500	(5, 6, 10)	+0.286	+0.143	+0.143	0.536
11	C	0.500	+0.500	(7, 8, 9, 11, 12)	−1.111	−0.556	−0.556	0.365
12	C	0.500	+0.500	(7, 8, 9, 11, 12)	−1.111	−0.556	−0.556	0.365

Log loss 0.693 → 0.515 (Part 3 stump: 0.504). Only run 10 is wrong ($p = 0.536$, it is **C**): it shares a leaf with runs 5 and 6.

Slide 92 — Only run 10 wrong (p = 0.536).

Slide 93 Rounds 1 to 6

Rounds 1 to 6

round	questions asked (root, then yes-side)	gains before γ	leaf weights w	loss	wrong
1	rate $\leq$ 0.065, then batch $\leq$ 48	1.667, 0.794	+0.29, −1.11, +1.00	0.515	10
2	rate $\leq$ 0.065, then batch $\leq$ 48	0.939, 0.462	+0.23, −0.84, +0.78	0.412	10
3	rate $\leq$ 0.065, then warm-up = no	0.575, 0.343	+0.17, −0.75, +0.64	0.344	none
4	rate $\leq$ 0.065, then batch $\leq$ 48	0.379, 0.282	+0.21, −0.63, +0.54	0.295	none
5	rate $\leq$ 0.065	0.256	−0.26, +0.46	0.276	none
6	no split	all below γ	+0.052	0.276	none

Training error is 0 from round 3. In round 5 only the root clears γ; from round 6 on no question gains more than 0.25, so the tree is a single leaf that shifts every score by only 0.03, 0.02, 0.01 and the loss stays near 0.275.

Slide 93 — Loss stalls near 0.275 once nothing clears γ (A6).

Slide 94 What λ and γ do

What λ and γ do

Same twelve runs, two-question trees, $\nu = 0.5$, eight rounds:

settings	first round with no wrong run	wrong after round 8	log loss after 8	leaves in trees 1 to 8
$\lambda = 0,\ \gamma = 0$	2	none	0.049	3, 4, 4, 4, 4, 4, 4, 4
$\lambda = 1,\ \gamma = 0$	3	none	0.190	3, 3, 3, 3, 3, 3, 4, 3
$\lambda = 1,\ \gamma = 0.25$	3	none	0.275	3, 3, 3, 3, 2, 1, 1, 1
$\lambda = 1,\ \gamma = 0.5$	never	10	0.369	3, 2, 3, 1, 1, 1, 1, 1
$\lambda = 1,\ \gamma = 1$	never	5, 6	0.573	2, 1, 1, 1, 1, 1, 1, 1

Larger λ and γ give smaller leaf values and smaller trees: the training loss falls more slowly (or stops), which is the intended price of a simpler model. Too large a γ (≥ 0.5 here) stops learning before the training data are fitted; choose both on held-out data.

Slide 94 — Choose both on held-out data.

Slide 95 Runs A and B through the trees

Runs A and B through the trees

round	run A (0.05, 32, no): path	νw	run B (0.2, 64, yes): path	νw
1	rate $\leq$ 0.065 yes; batch $\leq$ 48 yes	+0.143	rate $\leq$ 0.065 no	+0.500
2	rate $\leq$ 0.065 yes; batch $\leq$ 48 yes	+0.113	rate $\leq$ 0.065 no	+0.389
3	rate $\leq$ 0.065 yes; warm-up $=$ no yes	+0.086	rate $\leq$ 0.065 no	+0.319
4	rate $\leq$ 0.065 yes; batch $\leq$ 48 yes	+0.104	rate $\leq$ 0.065 no	+0.269
5	rate $\leq$ 0.065 yes	−0.131	rate $\leq$ 0.065 no	+0.232

Entry $= \nu w$ of the leaf reached. Rounds 6–8 add +0.026, +0.017, +0.012 to every run. Totals after 8 rounds: A $F = 0.369$, $p = 0.591$; B $F = 1.764$, $p = 0.854$. Both predicted **D**; B with more confidence. Unlike AdaBoost, XGBoost gives a probability.

Slide 95 — XGBoost gives a probability (A7).

Slide 96 XGBoost: summary of the procedure

XGBoost: summary of the procedure

step	what you do	which number is used
start	$F_i = 0$, $p_i = 0.5$; choose λ, γ, ν, depth	the three settings
(a) slopes	$g_i = p_i - y_i$, $h_i = p_i(1 - p_i)$	each run's current p
(b) question	largest gain $\frac{1}{2}[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}] - \gamma$	G, H of the node, yes, no
(c) grow	split again while the gain exceeds 0; stop at depth limit	the gain after γ
(d) leaf weights	$w = -G/(H + \lambda)$	G, H of the leaf's runs and λ
(e) update	$F_i \leftarrow F_i + \nu w$	w of the leaf reached
predict	$F = \nu \sum_t w_t(\text{leaf})$, $p = S(F)$	one weight per tree

Slide 96: the whole XGBoost procedure as a table, from slopes and curvatures to the gain, the leaf weights and the update.

Slide 97 Part 6: Comparison (divider)

One tree, many trees averaged, many small trees added.

Slide 98 Comparison of bagging and boosting

Comparison of bagging and boosting

	bagging (random forest)	boosting
trees grown	side by side, each on its own bootstrap sample	one after another, each on the residuals so far
each tree	deep: low bias, high variance	small (a stump): high bias, low variance
combined by	average $\frac{1}{B}\sum_b T_b$	sum $F_0 + \nu \sum_t \gamma_t(\cdot)$
mainly reduces	variance, down to the floor $\rho\sigma^2$	bias, step by step
more trees	error settles; harmless	keeps fitting; stop using held-out data
main settings	number of trees; features per split	step size; rounds; depth

Slide 98: bagging and boosting compared on how trees are grown, how deep they are, how they are combined and what they mainly reduce.

Slide 99 Gradient boosting, AdaBoost and XGBoost side by side

Gradient boosting, AdaBoost and XGBoost side by side

	gradient boosting (Part 3)	AdaBoost (Part 4)	XGBoost (Part 5)
each stump or tree is fitted to	residuals $y - p$	the data with run weights	slopes g and curvatures h
loss reduced	log loss	exponential loss e^{-yF}	log loss, plus penalties
question chosen by	$n_L \bar{r}_L^2 + n_R \bar{r}_R^2$	smallest weighted error ε	largest gain with $G^2/(H+\lambda)$
number each tree contributes	$\nu\gamma_{\text{leaf}}$	$\alpha\, h(x) = \pm\alpha$	νw_{leaf}
output	probability $S(F)$	class and margin F	probability $S(F)$
controls on size	step size, rounds	rounds	λ, γ, depth, step size

Trees, bagging, boosting ES 335 / ES 678 Machine Learning 99 / 102

Slide 99: gradient boosting, AdaBoost and XGBoost side by side, on what each tree fits, the loss reduced and how the question is chosen.

Slide 100 Predictions for the two new runs

Predictions for the two new runs

method	run A (0.05, 32, no)	run B (0.2, 64, yes)
single tree	**D** (2/2 in its leaf)	**D** (4/4 in its leaf)
bagging, 3 trees	0 1 1 $\to \frac{2}{3}$, **D**	1 0 1 $\to \frac{2}{3}$, **D**
bagging, many trees (simulated)	≈ 0.872, **D**	≈ 0.934, **D**
boosting, 8 stumps	$F = 1.069$, $p = 0.744$, **D**	$F = 1.490$, $p = 0.816$, **D**
AdaBoost, 5 stumps	$F = 0.672$ (margin), **D**	$F = 0.601$ (margin), **D**
XGBoost, 8 trees	$F = 0.369$, $p = 0.591$, **D**	$F = 1.764$, $p = 0.854$, **D**

The single tree gives only a class; bagging shows how much the trees disagree (with many trees, about 0.9 of them say **D**); boosting gives a probability built from summed corrections.

Trees, bagging, boosting ES 335 / ES 678 Machine Learning 100 / 102

Slide 100: every method's prediction for the two new runs A and B, each with its score or vote.

Slides 101–102 Review questions

Answered in Part C (Q1–Q11).

Also worth knowing

XGBoost for regression. With loss $\frac{1}{2}(F-y)^2$, the slope is $g = F - y$ and the curvature is $h = 1$. Take four points $x = 1, 2, 3, 4$, $y = 3, 5, 9, 11$, with $\lambda = 1$, $\gamma = 1$ and step $\frac{1}{2}$. Start at $F_0 = 7$, so $g = 4, 2, -2, -4$. For the split $x \le 2$, $G_L = 6$ and $G_R = -6$, with $H = 2$ on each side, so each leaf scores $36/3 = 12$. The parent scores 0, so the gain is $\frac{1}{2}(12 + 12 - 0) - 1 = 11$, the largest of the three splits. The leaf weights are $-6/3 = -2$ and $+2$, giving $F_1 = 6, 6, 8, 8$, and the total squared error falls from 40 to 20.

Part C Practice

Deck review (slides 101–102):

1. Gain of "batch ≤ 48" at the root (entropy)?
2. Which number do we read from bagged tree 2 for run B, and why?
3. σ² = 1, ρ = 0.3, B = 20: variance of the average and the floor?
4. Why does the bagging variance table match σ²/B but show no floor?
5. In boosting round 1, why is $\gamma_{\text{no}} = +2$ but each score rises by 1?
6. After round 3 two runs are wrong though the loss fell: contradiction?
7. A stump has weighted error 0.2. Vote weight? Its weighted error after the update?
8. Why are AdaBoost run weights called exponential-loss residuals?
9. What does λ do to a leaf value, and what does γ do to a tree?
10. With λ = 0, γ = 0 and stumps, which part of XGBoost still differs from L12c?

Extra: 11. A leaf has $G = -3$, $H = 0.5$. Compute w for λ = 0, 1, 5. Which leaves are hurt most by λ? 12. Node $(G, H) = (0, 3)$, λ = 1, split into $(-2, 1)$ and $(2, 2)$. Gain before γ? Largest γ that still keeps it? 13. Why did the no-group of round 1 (runs 1–4, $G = -2$, $H = 1$) refuse to split even with γ = 0? Show any split of it has negative gain. 14. Which method would you choose, and why: (a) you need calibrated probabilities and the best accuracy on tabular data; (b) you want a robust model with almost no tuning and a free error estimate; (c) you need a model a doctor can read?

1. $1 - \frac{4}{12}(0.8113) - \frac{8}{12}(0.9544) = 0.0933$.
2. $T_2(B) = 0$: warm-up is yes, so B goes to the leaf "4 draws, all C".
3. $0.3 + 0.7/20 = 0.335$; floor 0.3.
4. Only the resampling varies (one fixed dataset); the floor comes from the dataset itself varying.
5. γ is the full Newton step; the update multiplies by ν = 0.5.
6. No: each round lowers the loss, not the error count.
7. $\frac{1}{2}\ln 4 = 0.693$; afterwards exactly ½ ⇒ vote 0.
8. $w_i \propto e^{-y_i F_i} = |dL/dF|$ for $L = e^{-yF}$.
9. λ shrinks $w = -G/(H + \lambda)$ towards 0; γ blocks splits whose gain is below γ (smaller trees).
10. The question score: G^2/H (curvature-weighted) instead of $n\bar{r}^2$. (Equal only while all h are equal, e.g. round 1.)

11. λ = 0: $+6$; λ = 1: $+2$; λ = 5: $+0.545$. Small leaves (small H) are shrunk most — exactly the leaves most likely to be fitting noise.
12. $\frac{1}{2}[4/2 + 4/3 - 0/4] = \frac{1}{2}(3.333) = 1.667$; kept for any γ < 1.667.
13. Parent term $4/2 = 2$. All four runs have $g = -0.5$, $h = 0.25$; a split into k and $4-k$ runs gives $\frac{(0.5k)^2}{0.25k+1} + \frac{(0.5(4-k))^2}{0.25(4-k)+1}$. For $k = 2$: $2 \times \frac{1}{1.5} = 1.333$ ⇒ gain $\frac{1}{2}(1.333 - 2) = -0.333$; for $k = 1$: $\frac{0.25}{1.25} + \frac{2.25}{1.75} = 1.486$ ⇒ gain −0.257 (the slide's value). Splitting a group that already agrees only adds λ penalties.
14. (a) Gradient boosting / XGBoost, tuned with early stopping on held-out data. (b) Random forest: more trees never hurt, few settings, OOB error for free. (c) A single pruned tree (depth chosen by cross-validation) — readable, though unstable and less accurate.

Lecture 12 complete. Next: L12f — Supplement: syllabus topics the lectures did not cover (leakage, shift, validation design, optimization, multiclass, trees II, KNN). Back to the start: L12a — Decision trees.